

BMEG3105 Data analytics for personalized genomics and precision medicine

Scribing

Lecture 8: ‘Clustering’

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Survey Results:

Category	Strengths	Areas for improvement
Lecture Content and Delivery	- Content is clear, well-structured, and easy to follow - Topics are engaging and relevant, with repeated mentions of usefulness and interest - Overall quality rated highly (e.g., “Great!”, “Very good lecture”)	- Include additional examples for each measurement type to enhance understanding - Increase detail in slides to better support self-study - Provide introductory explanations for unfamiliar measurement types to improve accessibility
Tutorial Content and Delivery	- Tutorial notes and code explanations are clear and effective	- Promote greater interaction and dynamic engagement during tutorials - Add more practice questions to support skill development

Recap from last lecture -Clustering

1. Why Clustering?

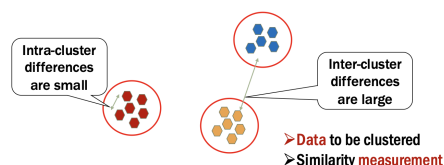
Cluster items: Improve organization and enable faster searching.

Cluster people: Tailor treatments for different patient groups (e.g., children, elders) and meet diverse customer needs by optimizing products, not just based on age or gender.

2. What is clustering?

Definition: Clustering is about finding groups of objects such that:

- Objects within the same group are similar (low intra-cluster difference)
- Objects in different groups are dissimilar (high inter-cluster difference)



Components of clustering:

- Data to be clustered
- Similarity/dissimilarity measurement (distance metric)
- Clustering algorithm (procedure for forming clusters)

3. How to do clustering?

Similarity and dissimilarity measurement

Similarity	Def: Numerical measure of how alike two data objects are - Higher when objects are more alike - Often falls in the range [0,1]
Dissimilarity (distance)	Def: Numerical measure of how different two data objects are - Lower when objects are more alike - Minimum dissimilarity is often 0 - Upper limit varies

Minkowski distance

Def: Generalization of Euclidean Distance

General Form: $dist(p, q) = (\sum_{k=1}^m |p_k - q_k|^r)^{\frac{1}{r}}$

$r = 1 \rightarrow$ City block/Manhattan distance (L1 norm) distance

Example: Hamming distance for binary vectors

$r = 2 \rightarrow$ Euclidean distance (L2 norm)

$r \rightarrow \infty$. “supremum” (Lmax norm, L_∞ norm) distance.

-Maximum difference between any component of the vectors

- r is a parameter
- m is no. of dimensions
- $p_k / q_k \rightarrow$ k-th attributes (components) or data objects p and q

Minkowski distance: Examples

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_∞	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Distance Matrix

New Topic

1. Hierarchical clustering
2. Mahalanobis distance (Advanced topic)

Hierarchical clustering

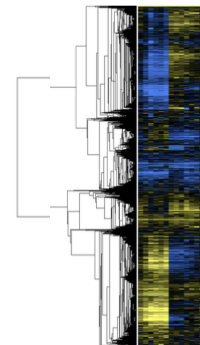
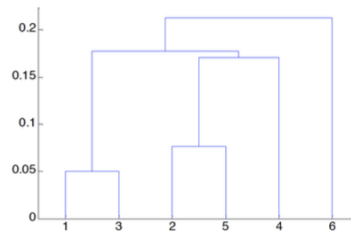
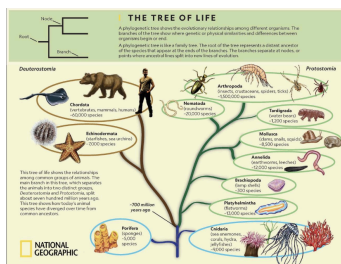
Key pt: ❖ Produces a set of nested clusters organized as a hierarchical tree

❖ Can be visualized as a dendrogram

-A tree like diagram that records the sequences of merges

❖ They may correspond to meaningful taxonomies

-Gene clusters, phylogeny reconstruction, animal kingdom



Steps of hierarchical clustering

1. Compute similarity or distance matrix.
2. Start: each data point = one cluster.
3. Merge the two closest clusters.
4. Update the similarity/distance matrix.
5. Repeat until only one cluster remains

How to update the distance matrix after merging?

- Min (single linkage): closest points between two clusters.
- Max (complete linkage): farthest points.
- Group average.
- Centroid distance

Example: GENE Expression Data

Gene	wt	mutant_1	mutant_2	mutant_3
At4g35770	1.5	3	3	1.5
At1g30720	4	7.5	7.5	5
At4g27450	1.5	1	1	1.5
At2g34930	10	25	23	15
At2g05540	1	1	2	1

Step 1: Use correlation as distance measure.

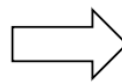
$$\rho_{X,Y} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

Step 2: Start with each gene as a separate cluster.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770	1				
At1g30720	0.9733	1			
At4g27450	-1	-0.9733	1		
At2g34930	0.9493	0.9909	-0.9493	1	
At2g05540	0.5774	0.562	-0.5774	0.4528	1

Step 3: Merge most correlated ones first.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9493	0.9909	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528	



... Until only a single cluster remains

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9493	0.9909	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528	

Step 4: Update distance matrix each time.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9733		-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.562	

Continue until one cluster remains

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-0.5774	-0.5774			
At2g34930			-0.5774		
At2g05540			-0.5774		

About programming

Python implementation:

Use scikit-learn (<https://scikit-learn.org/stable/>)

Example : Classification, Regression, Clustering

*Hierarchical clustering in Python

(https://scikit-learn.org/stable/auto_examples/cluster/plot_agglomerative_dendrogram.html#sphx-glr-auto-examples-cluster-plot-agglomerative-dendrogram-py)

Potential Project Application

Data preprocessing for gene expression matrix:

- Data collection and merging
- Exploration & visualization
- Data cleaning
- Get distance matrix
- Perform clustering

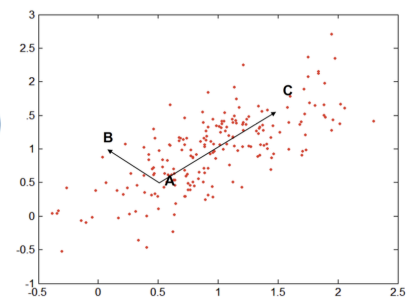
Advanced Topic - Mahalanobis distance

Key point: Calculating distance considering the data distribution

General Equation: $mahalanobis(p, q) = (p - q)^T \Sigma^{-1} (p - q)$

(Σ is the covariance matrix)

How to calculate the inverse of the covariance matrix?



$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\underset{\substack{\uparrow \\ \text{determinant}}}{ad-bc}} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} &= \frac{1}{4 \times 6 - 7 \times 2} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} \end{aligned}$$

Example:

❖ **Suppose we have two quizzes**

➤ **Quiz 1: std=10**

- Student A-student B=1

➤ **Quiz 2: std=1**

- Student C-student D=1

Same raw difference, but in standard deviation units, Quiz 2's gap is more significant

→ Normalization

- Standardizes scales → accounts for variance
- Data often needs normalization
 - (e.g., min-max) before applying distance metrics

Person	Height (m)	Weight (kg)
P1	1.79	75
P2	1.64	54
P3	1.70	63
P4	1.88	78

↓ Min-max normalization

Person	Height	Weight
P1	0.625	0.875
P2	0	0
P3	0.25	0.375
P4	1	1

Resource and Uncovered Topics

- Introduction to data mining: Chapter 2.4 & Chapter 8
- K-means clustering
- Density-based clustering
- How to determine the number of clusters
- How good is your clustering (lecture 8-9)